Swarm Field Problem Statement Report

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1 Reason for Problem Choice

Mapping individual plant data in a crop field enables early, targeted intervention against health risks like pests, reducing chemical usage and increasing crop yield. While per-plant health mapping provides clear benefits, relying on a single robot or a centrally controlled fleet creates operational bottlenecks, specifically regarding communication range and bandwidth limitations. For this reason, we decided to investigate the feasibility of mapping individual plant data using a decentralized swarm of robots.

2 Purpose of the Study

This study explores how swarms of robots can automate the cultivation and monitoring of crops. The aim is to design an algorithm in which each robot follows simple local rules and relies on its own knowledge, so that swarm behaviour emerges and the robots together form a collective brain: a shared body of knowledge about the field that no single robot holds on its own. The algorithm will be built with three global objectives in mind: keeping plants healthy, exploring the entire field, and achieving a high crop yield. To benchmark it, a virtual crop field will be modelled so that the behaviour of the swarm can be simulated over time.

3 Business Context

The client is the TU Delft Robotics Institute, a research entity whose mandate is to develop new technology. This shapes the project decisively: the problem sits at the start of the development chain, with no legacy system to extend and no platform to integrate with. The work begins from scratch, and the deliverable is validated knowledge and a proof of concept.

The strategic context is geopolitical change combined with the green transition. Food security rests on two things that are eroding at once: stable crop output, which climate instability disrupts, and available labour, which workforce-limiting events restrict. COVID-19 showed the latter sharply, when border closures stranded the seasonal migrant workforce western European agriculture depends on. The domestic stake is considerable, with Dutch agricultural exports reaching € 137.5 billion in 2025, a tenth consecutive year of growth [1].

A second context is small-scale, novel farming. Nettle grown for textiles is one example: a low-input perennial drawing interest as an alternative to cotton and synthetics. Such farms are small and each is different, so machines built for monoculture do not suit them. Robots come in many forms, drones, walkers, wheeled platforms, each suited to a different situation, so a swarm can adapt to many individual cases rather than one standard case. Industry has shown interest, and the coordination layer we develop is robot-agnostic, so it transfers across platforms.

4 Stakeholder Analysis & Value Proposition

Stakeholders

- **Farmers:** Need reliable information on crop health and pest distribution and would be the primary end-users of the data the swarm produces. Their adoption depends on cost, reliability, ease of use, impact on yield, and reduction in pesticide use making them the stakeholder with the highest influence on the design requirements.
- **Current and future field workers:** Directly affected by the introduction of the swarm, as tasks may shift from manual crop inspection to monitoring and maintaining the robotic system. This may require retraining or a change in role, making their acceptance important for successful adoption.
- **TU Delft / Robotics Institute:** Provides the resources, and academic coaching. Primarily interested in the research contribution and technical feasibility of the swarm behaviour, with high influence on the academic quality of the project.
- **Agricultural machinery, technology & robotic design companies:** Potential partners or competitors whose existing equipment and infrastructure the swarm concept must complement or integrate with, and who could support scaling the concept from prototype to a manufacturable product.
- **Environmental organisations:** Interested in whether the technology can meaningfully reduce pesticide use and environmental impact, and may act as advocates or critics depending on demonstrated results.

Value Proposition

We contribute a new concept for swarm robots in crop care, translating stakeholder needs especially farmers' and field workers' into simple local behavioural rules that scale to swarm-level monitoring. A simulation lets us tune these rules and inject noise to test robustness, so trade-offs (e.g. coverage speed vs. robustness, cost vs. reliability) can be explored before building a physical prototype, reducing risk and keeping the design grounded in real stakeholder priorities.

5 Sustainability Goals

- **SDG 2 – Zero Hunger:** Supports efficient and resilient crop production through automated monitoring.
- **SDG 9 – Industry, Innovation and Infrastructure:** Develops innovative decentralized swarm robotics for agriculture.
- **SDG 12 – Responsible Consumption and Production:** Enables targeted crop care, potentially reducing pesticide and resource use.

6 Problem Statement and Objectives

The objective of this study is to develop and explore decentralized swarm-level algorithm within ten weeks that enables a robot swarm to explore in a virtual crop field environment. The robot swarm will exhibit collective behaviour through simple local rules, potential-field-based navigation and peer-to-peer information sharing. The performance of the algorithm will be benchmarked based on predefined metrics to identify the most efficient approach for crop field monitoring.

7 Research/Design Questions and Scope

Sub-questions:

1. How to create a local decision engine that makes the collective swarm fulfil the global crop field objectives: exploration, monitoring and caring for plants needs.
2. How can the swarm understand the current needs of the crop field and react to them by rebalancing its composition through switching the robot bodies.
3. How do we make a swarm resilient to partitions, single robot outages, and uncertainty in collective swarm information?
4. How can the virtual crop field environment be used to benchmark the swarm and compare different decision and navigation algorithms?
5. What information is necessary to be shared, how often and between which robots to form a collective brain of the swarm?
6. What kinds of capabilities would the robot bodies need to have to be applicable to caring for a crop field, exploring, and monitoring it?
7. What is the difference between swarm head and swarm field?

Assumptions

1. There is no central control.
2. A robot can broadcast information to other robots within a limited range.
3. The crop field is of fixed-size.
4. An individual robot does not have enough storage to store the entire crop field map.

8 Project Plan / Approach

Week	Milestone / Activity	Deliverable
1–2	Problem framing, literature & expert consultation	Problem statement report
3–5	Exploring platforms and algorithm rules and implementing them	First working simulation
6	Initial experimentation and benchmarking	Midterm progress report
7–8	Implementing midterm feedback & Deadline Repo	Repository
9	Further experimentation and benchmarking	Final report
10	Preparing final presentation	Final review

References

[1] Statistics Netherlands (CBS) and Wageningen Economic Research (WUR). Value of agricultural exports up by over 8 percent in 2025. https://www.cbs.nl/en-gb/news/2026/03/value-of-agricultural-exports-up-by-over-8-percent-in-2025, jan 2026. Accessed: 2026-09-08.